

Arithmetic series



1 **a** $S_{10} = 195$ **b** $S_{10} = 0$ **c** $S_{10} = 280$ **d** $S_{10} = -310$

2 **a** $S_{15} = 450$ **b** $S_{15} = -45$ **c** $S_{15} = 105$

3 **a** $S_{10} = 47.5$ **b** $S_{10} = -53.75$ **c** $S_{10} = 85$ **d** $S_{10} = 5$

4 $S_{20} = 210\pi$

5 **a** $d = -8$ **b** $S_{14} = -798$

6 $S_{14} = \frac{273}{2}$

7 **a** $n = 18$ **b** $S_{18} = 450$

8 The sum of an arithmetic series is calculated as

$$S_n = \frac{n}{2} (2a_1 + (n - 1)d)$$

where n is the number of terms, a_1 is the first term and d is the difference between consecutive values.

Substitute $a_1 = 1$ and $d = 2$, we have:

$$\begin{aligned} S_n &= \frac{n}{2} (2 \times 1 + (n - 1) \times 2) \\ &= \frac{n}{2} (2 + 2n - 2) \\ &= \frac{n}{2} (2n) \\ &= n^2 \end{aligned}$$

9 **a** $S_{40} = 4920$ **b** $S_{40} = -2260$

10 $n = 10$

11 180



12 $\sum_{k=1}^{11} \left(-\frac{1}{2}\right)^{k-1}$